

Set **P**

**B.Sc. (E.C.S.) (Semester - II) (NEP CBCS) Examination:
March/April - 2026
Introduction to Microcontroller and Embedded System (G06-0203)**

Day & Date: Saturday, 11-04-2026
Time: 03:00 PM To 04:30 PM

Max. Marks: 30

Instructions: 1) All questions are compulsory.
2) Figures to the right indicate full marks.

Q.1 Choose correct alternative.

06

- Pin number 10 in 8051 microcontroller is _____.
 - RD
 - RXD
 - TXD
 - WR
- The 8051 Microcontrollers has _____ Timer.
 - 1
 - 2
 - 3
 - 4
- The output file generated by Keil for burning into 8051 is _____.
 - .hex
 - .bin
 - .obj
 - .exe
- 8051 Microcontroller Data Bus is _____ Bit.
 - 8
 - 16
 - 32
 - 64
- Before writing a program in Keil, the first step is _____.
 - Create a new project
 - Create a new Group
 - Select the microcontroller device
 - Write the code directly
- The internal ROM memory of the 8051 is _____.
 - 32 bytes
 - 64 bytes
 - 128 bytes
 - 4KB

Q.2 Answer the following. (Any Three)

06

- Define Hardware.
- Define Data transfer Instruction.
- Define Embedded System.
- Write note on Input output Ports.
- Define Register in 8051 Microcontroller.

- Q.3 Answer the following. (Any Two) 06**
- a) Explain difference between Microprocessor and Microcontroller.
 - b) Write down features of 8051 Microcontroller.
 - c) Write down any two characteristics of embedded system.
- Q.4 Answer the following. (Any Two) 06**
- a) Draw and explain Architecture of Embedded System.
 - b) Explain different types of Embedded System.
 - c) Write Embedded C program for LED blinking.
- Q.5 Answer the following. (Any One) 06**
- a) Draw and explain Block diagram of 8051 Microcontroller.
 - b) Explain with block diagram concept of Microcontroller based Embedded System.